

nRF93M1

Preliminary Datasheet

v0.7

nRF93M1 Feature overview

Features

Cellular

- Transceiver and baseband
- 3GPP LTE Release 14 Cat 1 bis compliant
 - Up to 5 Mbps uplink and 10 Mbps downlink
- RF transceiver for regional and global coverage
 - Power class 3, up to 23 dBm output power
 - Single 50 Ω antenna interface
- LTE band support in hardware
 - LABA: B1, B3, B5, B7, B8, B20, B28, B38, B40, B41
 - LACA: B1, B2, B3, B4, B5, B7, B8, B12, B13, B18, B19, B20, B25, B26, B28, B38, B40, B41, B66, B71
- Power saving features – DRX, eDRX, PSM
- IP v4/v6 stack
- SMS
- SIM, eSIM and softSIM support
 - Two ETSI TS 102 221 compatible UICC interfaces
 - SGP.02 M2M, SGP.22 Consumer and SGP.32 IoT

Wi-Fi beacon receiver for positioning

- IEEE 802.11b supported at 2.4 GHz band

Cellular location

- Cell-ID based location reporting

nRF Cloud Services

- Pre-provisioned for nRF Cloud services
 - Location observability and modem firmware update
- Seamless opt-in option for full range of host MCU services
 - Device management, firmware update, observability, and more

Peripherals

- Three UARTs
 - UART2 reserved as auxiliary channel
 - UART1, primary AT port
 - UART0, dedicated to modem trace
 - Autobauding and CTS/RTS hardware flow control
- USB 2.0 device interface
 - CDC-ACM, RNDIS, and CDC-ECM functions
 - Used for AT control, network data, and diagnostic trace
- Three pins for RF control interface
- Two pins for coexistence interface
- Module status and network status indicators

Integrated power management unit

- DC/DC regulators and low drop-out regulator (LDO) for internal power supply
- Power supply output for external components
- Voltage detectors for start-up, brown-out, and battery monitoring

Operating values

- -35°C to $+75^{\circ}\text{C}$ 3GPP-compliant operating temperature
- 3.3 V to 4.5 V supply voltage range, 3.8 V typical

Package

- 17.7×15.8 mm LGA (variable ball pitch), height 2.4 mm
- 109 pads, MSL 3

Applications

- Asset Tracking
- Fleet Management
- Security and Surveillance
- E-bikes and shared transport
- Gateway and Data Concentrators
- Telematics
- Point of Sales terminals
- Healthcare and remote patient monitoring

1 Revision history

Date	Version	Description
August 2026	0.7	Preliminary release

2 About this document

This document describes the nRF93M1 LTE Cat 1 bis cellular module: its features, electrical and mechanical characteristics, host integration interfaces, and operating conditions.

nRF93M1 ships pre-programmed with signed Nordic firmware that implements the LTE protocol stack, IP and upper-layer protocols, location features, and other functions. The host MCU controls the module through a documented AT command interface and standardized network interfaces; it does not execute customer code on the module itself. Firmware is updated in the field through Firmware Over-The-Air (FOTA) using HTTP/S, nRF Cloud service, or serial DFU.

The chapters in this document are organized by function, features and peripherals. Each chapter describes the feature, some relevant AT commands or signal interfaces, electrical characteristics where applicable, and any integration details or other relevant information.

2.1 Document status

The document status reflects the level of maturity of the document.

Document name	Description
Preliminary Datasheet	Applies to document versions up to 1.0. This document contains target specifications for product development.
Datasheet	Applies to document versions 1.0 and higher. This document contains final product specifications. Nordic Semiconductor ASA reserves the right to make changes at any time without notice in order to improve design and supply the best possible product.

Table 1: Defined document names

3 Product overview

nRF93M1 is an easy-to-use, compact, pre-certified LTE Cat1 bis module delivering global connectivity for more data-intensive cellular IoT. It supports up to 10 Mbps downlink and 5 Mbps uplink at low power in an optimized single antenna design. nRF93M1 ships preprogrammed with signed Nordic firmware that implements the LTE protocol stack, IP and upper-layer protocols, and other functions.

nRF93M1 is available in two variants. Both are pin, size, and software compatible, and share the same interfaces, AT command set, and firmware. They differ in supported LTE bands and in certification coverage.

- nRF93M1-LABA – Multi-regional LTE bands for Europe, Asia, the Middle East, and Africa, with corresponding regulatory and mobile network operator certifications.
- nRF93M1-LACA – Global comprehensive set of LTE bands and certifications for worldwide deployments.

Because the variants are pin and software compatible, you can develop against one and move to the other without a PCB redesign or application changes. Per-variant band lists are given in Supported bands and frequencies; order codes are given in Ordering information.

The integrated 2.4GHz Wi-Fi beacon receiver and cellular location enable indoor and outdoor positioning that are fully integrated with nRF Cloud location services. This provides an easy and power-efficient way for any device to obtain location information, while allowing the flexibility to add external GNSS depending on use case and performance requirements.

The standard, flexible peripherals focus on interfaces for secure, robust and optimized designs. These include both a 2-pin LTE coexistence interface and a dedicated 3-pin RF control interface allowing band-specific antenna tuner settings — recovering radiated efficiency across the full 700–2700 MHz band range for better link budget and lower power in compact, global, single-antenna designs.

Security is an integral part of nRF93M1 and nRF93M1 provides hardware-anchored device security, authenticated and encrypted firmware, and standards-based transport security (TLS 1.2, TLS 1.3, DTLS 1.2). nRF93M1 is designed to support regulatory security certification such as the CE RED radio equipment directive and the EU Cyber Resilience Act(CRA) cybersecurity regulation.

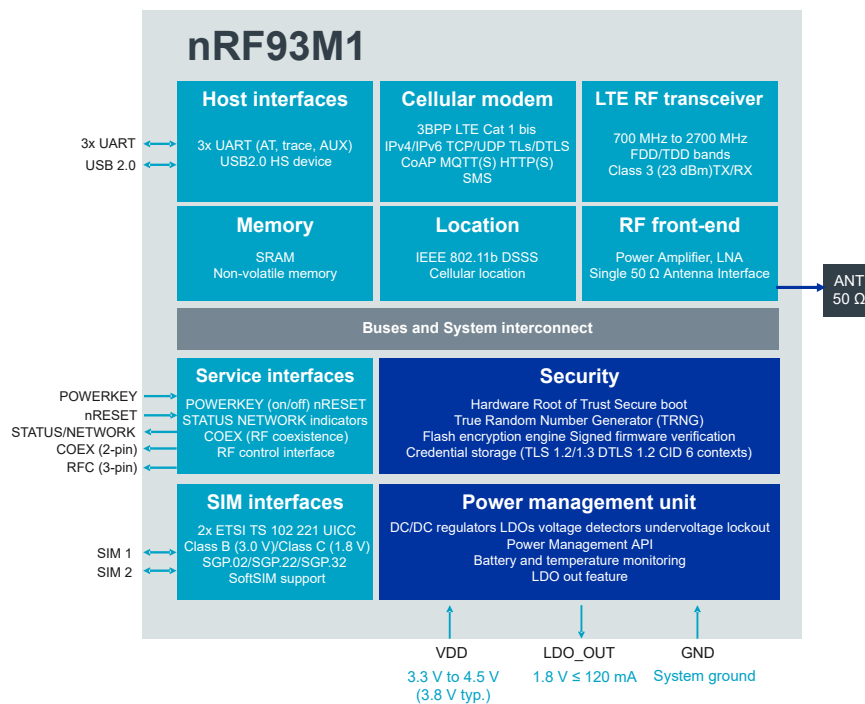
nRF93M1 comes with complete design and life cycle support also through Nordic's nRF54L host application support, which include complete application software and hardware reference designs. nRF93M1 also supports any other embedded or rich-OS host designs.

Nordic's AI-assisted development flow delivers concrete benefits when building wireless IoT solutions, including faster prototyping from idea to proof of concept on nRF93M1 Development Kit. Once your fleet of cellular IoT devices are deployed in the field, AI-assisted root-cause analysis and device debugging are enabled within the same development work flow.

nRF93M1 comes out of the box with nRF Cloud location, observability and modem-firmware-update services. It is further prepared to enable easy opt-in for a more complete set of life cycle services.

3.1 Block diagram

The following figure illustrates the overall system.



3.2 Feature matrix

Summary of the major features exposed on nRF93M1.

Cellular

Feature	Description
3GPP release	Release 14
LTE category	Cat 1 bis (FDD + TDD)
Power class	LTE Power Class 3 (23 dBm)
Supported LTE bands LABA	FDD – B1, B3, B5, B7, B8, B20, B28 TDD – B38, B40, B41
Supported LTE bands LACA	FDD – B1, B2, B3, B4, B5, B7, B8, B12, B13, B18, B19, B20, B25, B26, B28, B66, B71 TDD – B38, B40, B41
Peak data rate (DL / UL)	10/5 Mbps
Channel bandwidths	1.4 MHz, 3 MHz, 5 MHz, 10 MHz, 15 MHz, 20 MHz
Power saving	PSM (+CPSMS), eDRX (+CEDRXS), %PTWEDRXS
NAS protocols	IPv4, IPv6, IPv4v6, Non-IP
Simultaneous PDP contexts	4 (out of CID range 1–15)
SMS	PDU mode, 3GPP TS 27.005

Location

Feature	Description
Cellular-positioning data	Cell info and neighbor-cell data via %NRF_CLOUD_LOCATION
Wi-Fi scan for location	2.4 GHz scan via %WIFI_SCAN; results consumed by nRF Cloud location service

Host interfaces

Feature	Description
UART1 (primary AT port)	Up to 3 Mbps, autobaud, CTS/RTS
UART0 (trace)	Binary trace, 921,600–6,000,000 bps
UART2 (auxiliary)	Up to 3 Mbps
USB device	CDC-ACM, RNDIS, CDC-ECM composite
SIM	Single and dual SIM slot options, hot-swap detection, SoftSIM
Powerkey	Pin for graceful powering nRF93M1 up and down into System Disabled

Feature	Description
STATUS	Pin output for module status
NETWORK	Pin output for network status
Coexistence interface	Coexistence interface pins
Antenna tuner	RF control interface

Table 2: Other interfaces and features

nRF Cloud integration

Feature	Description
Transport	CoAP with DTLS 1.2 support
Device identity	%DEVICEUUID, %CLOUDACCESSKEY (P-256 ECC)
Authentication	JWT (%REGJWT, %JWT)
Services	Location, Messaging, Shadow, FOTA
Observability	Cloud core dump + metrics (%NRF_CLOUD_OBS)
FOTA transports	HTTP/S, nRF Cloud

Security

Feature	Description
Secure boot	Signed firmware verified against hardware root of trust
Transport security	TLS 1.2, TLS 1.3, DTLS 1.2 (Connection-ID)
Concurrent security contexts	6
Credential storage	File system: %FOPEN, %FWRITE, %FDELETE; referenced via %SSLCFG, %MQTTCFG

4 Power and clock management

nRF93M1 integrates a power management system that handles the voltage supply, system power-state, network-driven power-saving features, and reset behavior. Sleep states are managed through the system, delivering power efficiency without additional input from the host MCU.

The module has three system-wide power modes: System ON, System OFF, and System Disabled. Within these modes, the module continuously selects the most power-efficient internal configuration compatible with the current LTE modem state, the configured power-saving parameters, and the host interface activity without additional external input.

Power consumption follows the use model, not a host-selected state. The internal mechanisms by which the module achieves them are implementation-defined and may be improved in later firmware releases without changing these guarantees.

4.1 Power supply system

nRF93M1 is powered from a single external supply. An integrated power management unit (PMU) generates the rails required by internal domains, SIM interface, and I/O.

The host PCB provides a single power supply for the **VDD** pins and ties all **GND** pins to the system ground. Recommended decoupling and PCB layout are described in the *Hardware Design Guidelines*.

Pin	Direction	Description
VDD	Input	Module supply. 3.3 V to 4.5 V, 3.8 V typical. See Recommended Operating Conditions for more information.
LDO_OUT	Output	Internal LDO output available to the host, nominal 1.8 V, up to 120 mA. Also supplies the module's digital I/O pins internally.
SIM1_VOUT, SIM2_VOUT	Output	SIM-card supply, switched 1.8 V or 3.0 V depending on the inserted UICC class.

Table 3: Module-level supply pins

After VDD is applied, the module remains in System Disabled until the **POWERKEY** pin is asserted. The **POWERKEY** pin puts the module into System ON and back into System Disabled.

4.1.1 Power On and Off

nRF93M1 is powered on and off through the POWERKEY pin. It can also be powered off using the AT command `AT+POWD`.

POWERKEY pin

The POWERKEY pin has an always-on internal pull-up resistor that keeps it pulled high by default.

- In System Disabled mode, pull POWERKEY low for at least 10 ms to start the device and enter System ON.
- In System ON or System OFF mode, pull POWERKEY low for at least 650 ms to power down and enter System Disabled.

AT-command power and reset control

The host can also control the module power state through the AT command interface.

Command	Effect
AT%POWD	Power down the module to System Disabled. After this command, only POWERKEY wakes the module.
AT%RST	Software reset. The module restarts and re-enters System ON automatically.
AT%FACTORYRESET	Reset all user-configurable settings to default, then restart.
AT+CFUN=1, 1	Standard 3GPP TS 27.007 reset. Restarts the modem and module.

Table 4: AT-level power and reset control

4.1.2 I/O power supply

The digital I/O pins are driven from the module's internal LDO, fixed at 1.8 V. No external I/O supply is required from the host.

All standard digital pins (UART, status indicators, coexistence interface, and RF control interface) are referenced to the internal 1.8 V LDO output. The host PCB must use 1.8 V logic levels on these pins. See Recommended operating conditions for the I/O voltage tolerance.

The SIM interface pins follow the SIM_VOUT supply rather than the internal 1.8 V LDO, so the module supports both 1.8 V (Class C) and 3.0 V (Class B) UICC cards transparently. SIM_VOUT is auto-selected by the modem firmware based on the inserted UICC class.

4.1.3 LDO output

nRF93M1 generates regulated supply for external use as shown in the following table.

Pin	Voltage	Description
SIM1_VOUT SIM2_VOUT	1.8 V or 3.0 V (auto-selected per UICC class)	SIM card supply on the SIM_VOUT pin
LDO_OUT	1.8 V (nominal)	Supply, max 120 mA

Table 5: Output supply

4.2 Power supply and temperature monitoring

nRF93M1 continuously monitors input voltage and internal temperature, and reports threshold events to the host over the AT command interface.

Voltage monitor

Input voltage at the VDD pin and temperature are exposed to the host through the %ADC AT command.

- AT%ADC="vbat" returns the current input voltage in mV
- AT%ADC="temp" returns the current internal temperature in °C
- AT%ADC="all" returns both values

Power alarms

Alarms are received through %PALARM for the following:

- %PALARM: Volt,<voltAlarm> and %PALARM: Therm,<thermAlarm>
- Where <voltAlarm> - 0-1, Voltage alarm state. 0 means the voltage is below 2.8 V and 1 means voltage level has recovered.
- Where <thermAlarm> 0-1, Die temperature alarm state. 0 means the die temperature is below the threshold and 1 means temperature is above the threshold.

The alarm is triggered when one of the following thresholds are reached:

- Voltage drops below 2.8 V
- Die temperature is above 100°C

Note: %ADC and %PALARM monitoring are available in System ON active operation only. The ADC and monitoring domains are powered down during power-saving sleep.

An application that must monitor supply voltage or temperature continuously has to perform that monitoring from the host MCU.

Under-voltage lockout

If the **VDD** pin input voltage falls below the minimum operating voltage, the module transitions to minimum functionality. The host must issue +CFUN to restore full functionality. See the Recommended Operating Conditions chapter for supply voltage limits.

4.3 System power modes

The power management system transitions automatically between the defined operating modes based on configuration, host interaction, and the LTE modem state.

Mode	Description	Entry and exit
System ON	Full operation. The LTE radio, functional blocks, and peripherals are available. When power saving is enabled and the modem is in RRC Idle, eDRX, or PSM, the module sleeps automatically and wakes on host or network activity. Full application and network context, including sockets, TLS/DTLS sessions, and network registration, is preserved throughout.	Entered from System Disabled by POWERKEY held low for at least 10 ms. Power saving is enabled with %MODULECFG="autoPowerSave" and is automatically managed.
System OFF	Deep low-power sleep that retains network registration and all configured wake-up sources. SRAM is not retained and ongoing tasks are terminated. Opt-in only, and intended for long-cycle periodic operation. See 4.5.1.	AT%SYSOFF=1, which requires UART1 at a fixed baud rate of 9,600 bps or lower. Exit on any configured wake source, or AT %SYSOFF=0.
System Disabled	Lowest functional state. Nothing is retained and the module is deregistered from the network.	Default after VDD is applied. Entered with AT%POWD, or POWERKEY low for at least 650 ms. Wake by POWERKEY low for at least 10 ms, followed by a module boot.

Table 6: System wide power modes

4.3.1 System ON

In System ON, the LTE radio and all functional blocks are operational and managed by the modem firmware.

Within System ON, the modem firmware transitions automatically between active operation and System ON IDLE sleep to minimize current consumption while meeting LTE service requirements. Sleep transitions are reported to the host through the %SLEEPMODE URC.

The host application influences this behaviour through the commands in 4.6. No sleep-level selection is required or exposed.

4.3.2 System OFF

In System OFF, core system functionality is powered down and ongoing tasks are terminated. SRAM is not retained. Network registration is retained, and the module wakes from a defined set of sources.

Wake-up sources from System OFF:

- UART1 incoming data, if UART1 baud rate is configured to 9600 bps or lower before entering sleep.
- USB VBUS detect, wake when USB power is presented.
- POWERKEY signal pulled low.
- Internal paging timer, modem-internal wake on the next configured PSM or eDRX paging occasion.
- SIM hot-swap detect, wake when SIM presence transitions on a slot configured for detection.

Note: The System OFF configuration set by AT%SYSOFF=1 survives nRESET and a POWERKEY power cycle. After a reset the module returns to System OFF rather than to normal operation. Disable System OFF with AT%SYSOFF=0 before changing the UART configuration.

4.3.3 System Disabled

When VDD is applied, the module enters System Disabled. This is the lowest functional power state.

In System Disabled:

- The cellular modem and all peripherals are powered off.
- SRAM is not retained.
- The POWERKEY pin is the only wake source. Pull it low for at least 10 ms to enter System ON.

The host can place an active module into System Disabled using `AT%POWD`. After that command, only POWERKEY can wake the module again. The module performs a full boot.

4.4 Power saving

nRF93M1 supports the 3GPP power-saving mechanisms Power Saving Mode (PSM), extended Discontinuous Reception (eDRX), and idle and connected DRX, complemented by host-controllable modem functionality levels.

Power saving is largely transparent to the host. Once configured, the modem firmware manages state transitions automatically and the module enters the deepest power state compatible with the current connection and the traffic pattern. Host-side AT commands provide the configuration hooks.

4.4.1 Automatic power saving in System ON

Power saving within System ON is driven by the LTE modem state and is enabled by host configuration. Once enabled, no per-transaction host action is required.

- Power saving is enabled with `AT%MODULECFG="autoPowerSave", 1`. The setting can be changed only while the current sleep mode is System ON IDLE, meaning not while the module is already in power-saving sleep, and it takes effect when the radio is activated. The setting is not retained across a reset and must be re-applied.
- With power saving enabled, the module enters System ON power-saving sleep whenever the modem is in iDRX, eDRX, or PSM. It returns to full System ON automatically when the modem detaches, when the radio is turned off, or when there is data to send or receive.
- With power saving disabled, the module remains in System ON without sleeping. Disabled is the default at boot, and is the correct configuration for RF testing, provisioning, and any period in which peripheral or general-purpose I/O functionality must stay available.
- Power saving does not engage before the modem attaches. During network search, registration, and attach retries the module draws its full System ON current. Applications operating in marginal coverage should budget for this.

`AT%MODULECFG="slpWaitTime"` sets the time in ms for which the module stays awake after each wake-up, default 1 000 ms. Reduce this value to lower the average current on short duty cycles.

4.4.2 Host interface signaling DTR and RI

The UART1 DTR and RI pins let the host MCU and nRF93M1 sleep and wake independently, with no data loss and no additional GPIO.

- DTR is the host wake-up signal. Asserting DTR brings the module out of power-saving sleep so the host can transmit.
- RI is the module's data-ready signal. nRF93M1 asserts RI when it has data for the host, and de-asserts it once the module is ready to receive data from the host. RI can therefore serve as the host MCU wake-up source, allowing the host to sleep during periods of no traffic.
- nRF93M1 buffers incoming network data while the host wakes, so no data is lost even when the host wakes from a deep sleep state of its own.

- With hardware flow control, the host observes its CTS pin, driven by nRF93M1 UART1 RTS, to know when the module is ready to receive. A pull-up on the host CTS pin is required. Without hardware flow control, the host monitors RI, or applies a timeout of at least the module wake latency between asserting DTR and its first transmit.

DTR is the only host wake path that is independent of baud rate. Wake on incoming UART data requires UART1 to be at a fixed baud rate of 9600 bps or lower before sleep is entered. Designs that need to keep UART1 at 115,200 bps must use DTR to wake the module.

4.4.3 Power Saving Mode (PSM)

PSM is a 3GPP-defined low-power state in which the UE remains registered to the network but stops monitoring the paging channel, allowing long battery-saving sleep periods between user-initiated activity bursts.

PSM is requested through `AT+CPSMS`. The `<mode>` parameter controls whether PSM is enabled (1), disabled (0), or disabled with a parameter reset (2).

Network constraints

The network sets the granted timer values and may reject PSM entirely. Operator-specific minimums may also apply, such as a minimum T3412 of 190 minutes.

References

See 3GPP TS 23.682 for the PSM feature definition and 3GPP TS 24.008 for the timer encoding. The nRF93M1 AT Command Manual documents the full `+CPSMS` syntax.

4.4.4 Extended DRX (eDRX)

eDRX extends the paging cycle in the idle RRC state, trading response latency for lower average current on an otherwise idle cellular link.

eDRX is requested through `AT+CEDRXS`. The currently negotiated eDRX parameters are read back with `AT+CEDRXRDP`.

Paging Time Window

The proprietary `%PTWEDRXS` AT command configures the Paging Time Window (PTW) jointly with the requested eDRX parameters. PTW can be requested in the range 1.28 s to 20.48 s, and determines how long the UE monitors for paging within each eDRX cycle. Operator-specific minimums may also apply, such as a minimum of 5.12 seconds.

Relationship to PSM

eDRX and PSM are complementary and can be enabled simultaneously. The network and the application together decide the balance.

4.4.5 IDLE DRX and connected DRX

Discontinuous Reception (DRX) reduces receiver activity in the RRC the IDLE state. Connected DRX applies the same principle during an active RRC connection and is managed transparently by the modem.

IDLE DRX

In the RRC IDLE state, the modem wakes periodically at the shorter of the USIM-configured or the network-configured DRX cycle to monitor the paging channel. IDLE DRX parameters are set by the

network, and no host AT command changes them. Using eDRX extends the paging cycle beyond the short DRX cycles.

Connected DRX

Connected DRX is negotiated between the device and the network per the active RRC configuration and is handled by the modem. The modem firmware controls how System ON IDLE is used between the connected-DRX on-duration windows.

The host influences this decision through `%MODULECFG="pmuInCdrx"`:

- 0 — do not enter deep sleep during connected DRX
- 1 — allow deep sleep during connected DRX (default)

4.4.6 Modem functionality levels

`AT+CFUN` provides coarse-grained host control of the modem, allowing the LTE radio and stack to be deactivated without fully powering down the module.

<fun>	Behavior
0	Minimum functionality. RF TX off, LTE stack deactivated. The AT command interface and all non-radio functionality remain available.
1	Full functionality. RF on, LTE stack active.
4	Flight mode. RF TX off. All non-radio functionality remains available.

Table 7: Supported functionality levels

The `<rst>` parameter controls whether the command triggers a reset: 0 no reset (default), 1 reset and restart with normal-mode `+CFUN=1`.

Initial state at power-on

The device boots at `+CFUN=0` and does not attach automatically, giving the host full control of the attach sequence. The boot level is configurable through `%MODEMCFG="CfunInit", <fun>, <max_delay>:.`

- `<fun> = 0, 1, or 4` — initial functionality level.
- `<max_delay> = 0–65 535 s` — delay between modem boot and protocol stack power-on.

By default, the device uses minimum functionality (`<fun>=0`) with a `<max_delay>` of 0 s.

4.5 Host configuration

The following commands are the complete set required to select and configure the power modes and use models described in this chapter. No sleep-level configuration is required or exposed. Section 4.6.1 covers the additional settings that reduce current by disabling functions an application does not need.

Command	Purpose
AT+CFUN	Set modem functionality: 0 minimum, 1 full, 4 flight mode. The module boots at CFUN=0 and does not attach automatically, giving the host full control of the attach sequence. The boot level is configurable with %MODEMCFG="CfunInit".
AT+CPSMS	Request 3GPP Power Saving Mode timers T3324 and T3412-ext.
AT+CEDRXS, AT%PTWEDRXS	Request the extended DRX cycle and the Paging Time Window, 1.28 s to 20.48 s.
AT%MODULECFG="autoPowerSave"	Enable (1) or disable (0, default) power saving within System ON. Can be changed only while the current sleep mode is System ON IDLE. Takes effect on radio activation. Not retained across a reset.
AT%MODULECFG="slpWaitTime"	Time in ms for which the module stays awake after each wake-up. Default 1000 ms. Reduce this to lower the average current on short duty cycles.
AT%MODULECFG="pmuInCdrx"	Allow (1, default) or inhibit (0) power-saving sleep during connected DRX.
AT%SYSOFF	Enable or disable System OFF. Requires UART1 at a fixed baud rate of 9600 bps or lower.
AT%POWD	Enter System Disabled.

Table 8: Power management AT commands

4.5.1 Recommended power saving configuration

The commands in Table 4-9 select the power mode and the use model. The settings below reduce current further by disabling functions that an application in the field does not need. Apply them once during provisioning, and verify each returns OK before continuing.

```

AT+IPR=9600                // use LPUART
OK
<= change terminal software to 9600 bps =>
AT%MODULECFG="ledMode",0    // disable LEDs
OK
AT%MODULECFG="logCtrl",0    // disable log output
OK
AT%MODULECFG="usbCtrl",2    // disable USB
OK
AT%MODULECFG="usbSlpMask",1 // mask USB voting
OK
AT%SIMCFG="SimPowerSave",1  // enable SIM power save
OK
AT%SIMCFG="SimPresenceDetect",0 // disable SIM detection
OK

```

Order matters for the first command. AT+IPR=9600 takes effect immediately, so the terminal or host UART must change to 9 600 bps before the remaining commands are sent.

Command	Effect
AT+IPR=9600	Set UART1 baud rate to 9600 bps, which enables low-power UART operation and makes UART incoming data available as a wake source. Required for System OFF. See 4.4.2 for the alternative when the design must keep UART1 at 115,200 bps.
AT%MODULECFG="ledMode",0	Disable the STATUS and NETWORK indicator outputs. See 8.1.
AT%MODULECFG="logCtrl",0	Disable modem trace output. See 10.
AT%MODULECFG="usbCtrl",2	Disable the USB device interface.
AT%MODULECFG="usbSlpMask",1	Mask USB from the sleep vote, so an attached USB host does not hold the module awake.
AT%SIMCFG="SimPowerSave",1	Enable SIM power saving.
AT%SIMCFG="SimPresenceDetect",0	Disable SIM presence detection.

Table 9: Recommended power saving settings

"SimPresenceDetect", 0 disables SIM hot-swap detection, so SIM presence transitions no longer wake the module. Leave presence detection enabled on designs with a user-accessible SIM slot that must respond to a card being inserted or removed.

"usbCtrl", 2 disables the USB device interface. Designs that rely on USB VBUS detect as a wake source, or that use USB for AT control, network data, or trace, should leave USB enabled and use usbSlpMask alone to prevent USB from holding the module awake.

Note: Disabling trace output with "logCtrl", 0 also disables the diagnostic stream that Nordic technical support uses. Re-enable it before capturing a trace for a support case.

4.6 Reset

nRF93M1 can be reset through the dedicated nRESET pin or through AT commands.

nRESET pin

nRESET is active-low and has an internal 120 kΩ pull-up, so it remains high if left floating. The host PCB does not need an external pull-up resistor.

- Pulling nRESET low for at least 10 ms and then releasing it triggers a module reset.
- nRESET is intended for emergency or production-line reset only. In normal operation, prefer the AT-level reset commands.
- The pin may be left floating in production designs that do not use it.

AT reset commands

Command	Effect
AT+RST	Software reset. The module restarts and re-enters System ON.
AT+CFUN=1, 1	3GPP TS 27.007 reset. Restarts the modem.
AT+FACTORYRESET	Reset all user-configurable settings to factory defaults, then restart.
AT+POWD	Enter System Disabled. Only POWERKEY can wake the module afterwards.

Note: A reset does not clear the System OFF configuration. See the warning in 4.3.2.

4.7 Current consumption

Typical current consumption figures for nRF93M1 by operating state.

All values are measured at the module VDD pins with a conducted 50 Ω load on the antenna port. Typ is at maximum TX power, VDD = 3.8 V, TA = 25°C. PSM, eDRX, and DRX were measured in System OFF. Worst-case and best-case values depend on network configuration and are not controlled by the module.

State	Condition	Typ	Unit
System Disabled	AT+POWD	0.5	μ A
PSM floor	T3412 = 1 h	2.2	μ A
PSM cycle average	T3412 = 1 h, T3324 = 60 s, DRX = 0.64 s	25	μ A
eDRX	eDRX = 81.92 s, PTW = 5.12 s, DRX = 0.64 s	120	μ A
eDRX	eDRX = 655.36 s, PTW = 5.12 s, DRX = 0.64 s	25	μ A
Idle DRX	DRX = 0.64 s, USB1 unplugged	650	μ A
Idle DRX	DRX = 0.64 s, USB1 plugged	330	μ A
Connected DRX	DRX = 0.64 s	4	mA
Minimum functionality	AT+CFUN=0	2.2	μ A
Flight mode	AT+CFUN=4	2.2	μ A
Wi-Fi scan	10 s	30	mA

Table 10: Current consumption nRF93M1-LABA

		B8 (typ.)	B28 (typ.)	B3 (typ.)	B7 (typ.)	B40 (typ.)	Units
I _{RMC_23DBM}	Pout 23 dBm, QPSK, 1RB, 5MHz, RMC settings as per 3GPP TS 36.521-1 Annex A.2	515	570	605	550	225	mA

Table 11: Average current consumption, Cat 1 bis

5 LTE Modem and connectivity stack

The LTE Cat 1 bis modem is 3GPP release 14 compliant and supports Class 3, up to 23dBm output over a single 50 Ω antenna pin on FDD and TDD bands. It is field upgradable through differential firmware updates over serial and cellular interfaces.

It is optimized for low power operation and hence also supports LTE power saving mechanisms such as power saving mode (PSM) and extended discontinuous reception (eDRX).

In addition to the standardized LTE L1-L3 stacks the modem also included IP and the most used higher layer connectivity stack and security layers.

- IPv4 and IPv6
- TCP and UDP
- TLS and DTLS
- CoAP
- MQTT
- HTTP(S)

Additional stack options can be supported in the future as part of the firmware update releases.

5.1 Supported bands and frequencies

nRF93M1 supports the LTE bands listed in the following table. Individual bands may be enabled or disabled at runtime via the `AT%BAND` command.

Band list

Band	Variant	Uplink (MHz)	Downlink (MHz)	Duplex
B1	LACA/LABA	1920–1980	2110–2170	FDD
B2	LACA	1850–1910	1930–1990	FDD
B3	LACA/LABA	1710–1785	1805–1880	FDD
B4	LACA	1710–1755	2110–2155	FDD
B5	LACA/LABA	824–849	869–894	FDD
B7	LACA/LABA	2500–2570	2620–2690	FDD
B8	LACA/LABA	880–915	925–960	FDD
B12	LACA	699–716	729–746	FDD
B13	LACA	777–787	746–756	FDD
B18	LACA	815–830	860–875	FDD
B19	LACA	830–845	875–890	FDD
B20	LACA/LABA	832–862	791–821	FDD
B25	LACA	1850–1915	1930–1995	FDD
B26	LACA	814–849	859–894	FDD
B28	LACA/LABA	703–748	758–803	FDD
B66	LACA	1710–1780	2110–2200	FDD
B71	LACA	663–698	617–652	FDD
B38	LACA/LABA	2570–2620	2570–2620	TDD
B40	LACA/LABA	2300–2400	2300–2400	TDD
B41	LACA/LABA	2496–2690	2496–2690	TDD

Runtime band selection

The set of enabled bands can be locked or restricted with `AT%BAND`. The command accepts a bitmap of bands to enable; `AT%BAND=?` returns the full supported-band list. See the AT Command Manual for the exact syntax.

Peak data rates

Direction	Peak rate
Downlink	Up to 10 Mbps
Uplink	Up to 5 Mbps

Achievable rates depend on channel conditions, the serving cell's configuration, and the application-layer protocol overhead.

5.2 Connectivity stacks

nRF93M1 provides a full IPv4/IPv6 data-plane with integrated TCP, UDP, TLS, DTLS, HTTP, HTTPS, MQTT, CoAP, and sockets, exposed to the host through AT commands and a USB network interface.

Transport and application protocols

Feature	Details
IP versions	IPv4, IPv6, IPv4v6 (dual-stack), Non-IP. PDP type configured via +CGDCONT.
Sockets	Up to 12 concurrent TCP/UDP sockets (%SKTCREATE... family). Per-send payload up to 1400 bytes. Bind, listen (backlog 1–4), accept, and Release Assistance Indication (RAI) on send.
DNS	%DNS=<url> resolves a domain name to an IP address.
ICMP ping	%PING — count 1–255 or continuous, payload 1 B to 1500 B, per-request timeout 1000 ms to 600 000 ms, and Release Assistance Indication (RAI) on send..
ICMP echo reply	Gated by %MODULECFG="icmpEchoReply" (0 disabled, 1 IPv4 and IPv6 (default), 2 IPv4 only, 3 IPv6 only).
HTTP/HTTPS client	GET, range-GET, POST, POST from file, READ, READ to file. URL up to 2048 bytes; file body up to available file-system space. Content-type and SSL context per %HTTPCFG.
CoAP	Used by the nRF Cloud services over DTLS 1.2 (libcoap).
TLS / DTLS	Mbed TLS 3.6.5; TLS 1.2, TLS 1.3, DTLS 1.2; DTLS Connection-ID (RFC 9146); 6 concurrent contexts. See the Security chapter for cipher suites.

5.3 SMS

The nRF93M1 supports SMS in PDU mode for mobile-originated and mobile-terminated short messages, with support for delivery-status reporting.

Operational mode

The public AT command set exposes **PDU mode only**. Encoding follows 3GPP TS 23.040 and TS 23.041.

SMS AT commands

Command	Purpose
+CMGS	Send an SMS (mobile-originated).
+CMT	Unsolicited result code for an incoming SMS delivered directly to the TE.
+CNMA	New-message acknowledgement to the network.
+CDS / +CDSI	Delivery-status URC: direct (+CDS) or stored-in-ME (+CDSI).
+CMTI	Unsolicited result code for an SMS stored in the module.
+CNMI	Subscribe to SMS URCs and select delivery routing.
+CSCA	Configure the SMS service-centre address.

Constraints

- Only one AT client at a time may be registered as the SMS client. Any new SMS registration requires the previous registration to be released.
- Cell broadcast (+CSCB) is **not supported**.

5.4 Firmware update

nRF93M1 supports two over-the-air firmware-update transports and UART or USB based production-line flashing path. Each over-the-air path delivers cryptographically signed images that are verified against the hardware root-of-trust before activation.

Over-the-air firmware update

Path	Description
HTTP(S) FOTA	AT%HTTPFOTADL downloads an image directly from a host-supplied URL. TLS security context, content-type, and transport parameters are set through %HTTPCFG.
nRF Cloud FOTA	AT%NRFCLOUDFOTA with operations: 0 check host-application OTA, 1 download host-application chunk, 2 check modem firmware, 3 check and apply modem firmware, 4 clear cached URL. Chunks are Base64-encoded with CRC-16/CCITT-FALSE framing.

Production-line firmware updates

For production-line and bring-up, the module accepts firmware over UART using the Nordic download tool. The download agent boots at 115 200 bps and can be set to 3 Mbps for faster transfer.

Security guarantees

All firmware images are cryptographically signed. The signature is verified during the boot stage against the hardware root-of-trust. Images that fail verification are rejected and the module rolls back to the previously known-good image. See the Security chapter for details.

5.5 AT command interface

nRF93M1 is controlled by the host through a standards-aligned AT command set, exposed on the primary UART and on a USB CDC-ACM interface.

Standards base

- ITU-T V.250 — generic AT command syntax, serial-line behaviour, baud, flow control.
- 3GPP TS 27.007 — general, network, and mobile-termination commands.
- 3GPP TS 27.005 — SMS AT commands (PDU mode).
- Nordic-specific extensions — prefixed with % (for example %MODULECFG, %NRFCLLOUD*).

Result codes

Final result codes emitted by the module are OK, ERROR, +CME ERROR: <err>, and +CMS ERROR: <err>.

Verbosity is selected by +CMEE: 0 disabled (generic ERROR), 1 numeric, 2 verbose. Vendor-specific +CME ERROR values start at 512.

Runtime command discovery

The list of AT commands available at runtime can be retrieved with AT+CLAC.

Multiplexing

The module supports 3GPP TS 27.010 multiplexing (CMUX) via AT+CMUX, in both basic (transparency 0) and advanced (transparency 1) variants. Port speeds 600 bps to 3 Mbps are supported. A typical invocation AT+CMUX=0 starts basic CMUX.

5.6 SIM Interfaces

Two external SIMs and software SIMs are supported with commands/API to support selection between the SIMs. SIM cards with ISO/IEC 7816-3 specified interfaces are supported and the supported UICC/eUICC interface is compliant with the following

- ETSI TS 102 221 – Smart Cards, UICC-Terminal interface, Physical and logical characteristics
- ETSI TS 103 383 – Smart Cards, Embedded UICC, Requirements Specification

The physical interface for the eUICC is the same as the removable UICC. Both class B (supply voltage 3.0 V) and class C (supply voltage 1.8 V) operation is supported. If two external SIMs are used, both must use the same class/voltage.

For eUICC, M2M SGP.02, Consumer SGP.22, and IoT SGP.32 SIM provisioning is supported.

SoftSIM is supported through a standard SIM API, available upon request.

5.7 RF control interface

nRF93M1 provides a dedicated 3-pin interface for controlling external RF applications, such as antenna tuner devices. See the *nRF93M1 Cellular AT Command* document for more information.

5.8 LTE coexistence interface

A dedicated two-pin interface for RF control and RF interference avoidance is available for companion radio devices such as an external positioning device or Bluetooth® Low Energy device.

See the *nRF93M1 Cellular AT Command* document for more information.

5.9 RF front-end interface

The device has a single-ended (SE) 50 Ω antenna interface where an antenna is connected.

5.10 RF electrical specification

TX power, RX sensitivity, and maximum input power at the nRF93M1 antenna port.

TX power

nRF93M1 operates at LTE Power Class 3 up to 23 dBm output power (nominal 23 dBm conducted).

RX sensitivity

Band	Channel BW	Min.	Typ.	Units
B1	5 MHz	-99.5	-102	dBm
B3	5 MHz	-97	-101	dBm
B5	5 MHz	-100.5	-102	dBm
B7	5 MHz	-99	-101	dBm
B8	5 MHz	-101.5	-103	dBm
B20	5 MHz	-100	-103	dBm
B28	5 MHz	-100	-103	dBm
B38	5 MHz	-100.5	-102.5	dBm
B40	5 MHz	-100.5	-102.5	dBm
B41	5 MHz	-100.5	-102.5	dBm

Table 12: LABA RX sensitivity

Production RF test support

For production-line RF characterization the nRF93M1 supports a non-signalling RF test mode via `AT%RFTEST`. The command supports single-tone transmit at 0–23 dBm and RSSI-based receive characterization across the supported bandwidth set (1.4, 3, 5, 10, 15, 20 MHz). The `AT%PRODDONE` command latches the production-complete flag after the test sequence.

6 Wi-Fi Location

nRF93M1 can operate as a Wi-Fi beacon receiver supporting IEEE 802.11b DSSS reception in the 2.4 GHz band for access point (AP) based positioning. It performs signal reception and passive detection and decoding of beacon frames. Operations can be time-multiplexed with cellular functionality.

Scanning occurs when LTE is in RRC Idle mode (DRX/eDRX), PSM, or when cellular functionality is fully deactivated and can be configured and started through AT Commands. Positioning is determined by a location service such as Nordic nRF Cloud using the reported beacon data.

This is a location solution ideal for cost and power sensitive devices. For more information also please refer to the nRF93M1 AT Command documentation

7 Cellular Location

Cellular location enables the estimation of device position based on the mobile network cells visible to the specific device. The nRF93M1 reports serving-cell and neighbour-cell information through AT commands. Positioning is determined by a location service such as Nordic nRF Cloud using the reported cell data.

This is a solution ideal for cost and power sensitive devices. For more information also please refer to the nRF93M1 AT Command documentation

8 Device interfaces

nRF93M1 exposes host-side and other interfaces for communication, data, control, and status.

8.1 Status and Network indicators

The nRF93M1 includes module and network status indicator features available through the STATUS and NETWORK pins.

8.1.1 STATUS indicator

The STATUS pin indicates the nRF93M1 operational status during all modes excluding System OFF.

Mode	Description
Pin is high	Module is powered, initialized and running normally
Pin is low	Module is not properly powered and/or not initialized and running normally

8.1.2 NETWORK indicator

The **NETWORK** pin indicates the nRF93M1 network status during all modes excluding System OFF.

Network status	Pin HIGH (ms)	Pin LOW (ms)	Description
Initializing	200	1800	Slow blink
Connected	1800	200	Mostly on
Disconnected	200	1800	Slow blink
Out of service	200	1800	Slow blink
Data transfer	125	125	Fast blink

8.2 USB

The nRF93M1 module has a USB 2.0 compliant interface that supports Full-Speed (12 Mbps) and High-Speed (480 Mbps) modes. This interface can be used for commands, data, and tracing.

8.3 UART

nRF93M1 provides three asynchronous UART peripherals, each with dedicated functionality:

- UART1 — primary AT command and data port.
- UART0 — modem trace output.

- UART2 — auxiliary data channels.

The UARTs support low-power operation and can be used as nRF93M1 wake-up sources from System OFF sleep when the UART is configured to a baud rate of 9600 bps or less before entering System OFF.

8.3.1 UART1 – Primary

UART1 is the primary serial interface used for AT commands, data, and firmware updates.

The default baud rate is 115,200 bps. The following baud rates are supported:

- 600 bps
- 1200 bps
- 2400 bps
- 4800 bps
- 9600 bps
- 19,200 bps
- 38,400 bps
- 57,600 bps
- 115,200 bps
- 230,400 bps
- 460,800 bps
- 921,600 bps
- 3,000,000 bps

Autobauding is enabled by setting `+IPR=0`. CTS/RTS hardware flow control is supported.

UART1 supports low-power operation and can be used as nRF93M1 wake-up source from System OFF when the UART is configured to 9 600 baud prior to entering System OFF. The DTR pin is the wake input.

8.3.2 UART0 – Trace

UART0 is dedicated to modem trace output from the nRF93M1. It carries a proprietary binary log stream for use with Nordic's cellular monitor tool or for capture and supply to Nordic as part of technical support.

The default baud rate is 3 Mbit/s and the port supports up to 6 Mbit/s.

See the Modem trace topic in the Debug and trace chapter for runtime port selection and trace configuration.

8.3.3 UART2 – Auxiliary

UART2 is an auxiliary serial interface available for applications preferring or needing a second data/communication channel.

9 Security

The nRF93M1 provides hardware-anchored device security, authenticated and encrypted firmware, and standards-based transport security (TLS 1.2, TLS 1.3, DTLS 1.2).

Platform security

The nRF93M1 platform security is anchored in module hardware and includes the following hardware-backed mechanisms:

- Secure boot. A hardware root-of-trust verifies the authenticity and integrity of every firmware image loaded at power-on. Unsigned or tampered images are rejected.
- Flash encryption. Code and secrets stored in internal non-volatile memory are protected by on-chip flash encryption, tying the stored image to the device.
- Hardware true random number generator (TRNG). An on-chip TRNG seeds all cryptographic operations (TLS handshake ephemerals, key generation, nonces).
- Signed firmware update. Firmware updates are cryptographically verified against the root-of-trust before activation; failed verification triggers rollback to the previously known-good image. See [Firmware update](#) on page 23 for supported firmware-update transports (HTTP and nRF Cloud).

Transport security

The nRF93M1 provides transport-layer security services built on Mbed TLS 3.6.5:

- TLS 1.2 and TLS 1.3 (RFC 5246, RFC 8446).
- DTLS 1.2 (RFC 6347), used for CoAP over UDP.
- DTLS Connection-ID per RFC 9146 for NAT resilience on power-saving cellular links.
- Up to 6 concurrent TLS contexts, each independently configurable for protocol, cipher suite, certificates, PSK identity, SNI, and session cache.
- Three security levels (`seclevel 0` — no authentication, `seclevel 1` — server authentication, `seclevel 2` — mutual authentication).

The cipher suites exposed through the AT command interface (`%SSLCFG`) are found in the *nRF93M1 AT Commands*.

Pre-shared-key (PSK) authentication is supported when both `psk_identity` and `psk_key` are configured on the same SSL context.

Credential storage

X.509 certificates, device private keys, and root CAs are stored as files in the module's file system. They are managed through the file system AT commands:

- `%FOPEN / %FWRITE / %FCLOSE` — write a PEM-encoded credential (CA certificate, device certificate, or private key) to a named file.
- `%FREAD` — read back a stored file.
- `%FDELETE` — delete a stored credential file.
- `%FSHA256` — verify the SHA-256 hash of a stored file.

TLS contexts reference credentials by file path using `%SSLCFG` (for example, `"cacert"`, `"clientcert"`, `"clientkey"`). Credentials are retained across reset and power cycles.

10 Debug and trace

The nRF93M1 supports firmware tracing through a dedicated UART or USB modem-trace port.

The nRF93M1 produces a real-time binary modem trace in the UniLog format, which can be captured over UART or USB for post-processing with the Nordic tool-chain.

Trace output

Trace level and output is selectable at runtime with the `%MODULECFG="logPortSel"` AT command and is produced on:

- UART0 at up to 6 Mbit/s (minimum 921 600 bit/s).
- USB diagnostic ACM port (CDC-ACM interface on the module's USB composite device).
- Either port individually, or both simultaneously.

The trace stream is live; no internal buffering or storage is required by the module itself. The host must capture the stream in real time to avoid drop.

Coredump and metrics export

For field-deployed devices, the nRF93M1 integrates nRF Cloud firmware-observability, which stores coredumps and periodic metrics heartbeats in the secure file system and uploads them to the nRF Cloud on demand. See the nRF Cloud services chapter for more information.

11 nRF Cloud services

nRF93M1 integrates with nRF Cloud to provide device provisioning, data messaging, device shadow, cell- and Wi-Fi-based location, firmware update, and firmware observability.

Overview

nRF Cloud services are exposed through a family of `%NRF_CLOUD*` AT commands. The transport is CoAP over DTLS 1.2 (libcoap and Mbed TLS), which keeps message overhead small while secure and is compatible with cellular power-saving modes. Each cloud transaction emits a `%COAP: URC` that carries the transport-layer status for that request.

Provisioning and authentication

Each device presents two stable identities and two short-lived credentials to the cloud:

- `%DEVICEUUID` — a 36-character device UUID that is generated once and stored in non-volatile memory.
- `%CLOUDACCESSKEY` — a P-256 ECC public key in Base64-DER form. The corresponding private key is generated on-chip once and never leaves the device.
- `%REGJWT` — a signed registration JWT used during device on-boarding.
- `%JWT` — a short-lived authentication JWT (approximately 1 hour validity). The device system time must be valid for the JWT to be accepted by the cloud.

Data services

- **Location** (`%NRF_CLOUDLOCATION`) — several location methods covering single-cell, multi-cell, Wi-Fi, and combinations. The response carries `<latitude>`, `<longitude>`, `<uncertainty_m>`, `<fulfilled_method>`. Multi-cell and Wi-Fi methods require the modem to be in RRC Idle.
- **Messaging** (`%NRF_CLOUDMESSAGE`) — publishes a JSON payload to the cloud over CoAP (for example `{"appId": "BUTTON", "data": "1"}`).
- **Shadow** (`%NRF_CLOUDSHADOW`) — reads a dotted path from the device's desired shadow and writes a dotted path into the reported shadow.
- **FOTA** (`%NRF_CLOUDFOTA`) — see the firmware-update description in the LTE Modem chapter.

12 Hardware and layout

12.1 Pin assignments

The following sections describe the nRF93M1 hardware and layout specifications.

12.1.1 Module pin assignments

The pin assignment table and figure describe the assignments for this variant of the chip.



Figure 1: Module pin assignments, top view

Pin	Name	Function	Description
1	GND	Power	Ground

Pin	Name	Function	Description
2	GND	Power	Ground
3	GND	Power	Ground
4	GND	Power	Ground
5	RESERVED		Connect thermally and mechanically to the application board but leave electrically unconnected.
6	RFC1	Digital I/O	RF control interface
7	POWERKEY	Module Power on/off	Power the module on or off
8	GND	Power	Ground
9	GND	Power	Ground
10	GND	Power	Ground
11	SIM1_IO	Digital I/O	SIM1 data
12	SIM1_RST	Digital I/O	SIM1 reset
13	SIM1_CLK	Digital I/O	SIM1 clock
14	SIM1_VOUT	Power	SIM power supply
15	nRESET	Digital I/O	Module reset pin
16	NETWORK	Digital I/O	Network status indicator
17	UART1_RXD	Digital I/O	UART1 Receive
18	UART1_TXD	Digital I/O	UART1 Transmit
19	UART1_DTR	Digital I/O	UART1 Data Terminal Ready
20	UART1_RI	Digital I/O	UART1 Ring Indicator
21	RESERVED		Connect thermally and mechanically to the application board but leave electrically unconnected.
22	UART1_RTS	Digital I/O	UART1 Request To Send
23	UART1_CTS	Digital I/O	UART1 Clear To Send
24	LDO_OUT	Power	LDO power out. nominal 1.8V, max 120mA
25	STATUS	Digital I/O	Module operating status
26	GND	Power	Ground
27	GND	Power	Ground
28	UART2_RXD	Digital I/O	UART2 Receive
29	UART2_TXD	Digital I/O	UART2 Transmit
30	GND	Power	Ground
31	GND	Power	Ground
32	GND	Power	Ground

Pin	Name	Function	Description
33	GND	Power	Ground
34	GND	Power	Ground
35	ANT	RF	Single-ended 50 Ω LTE antenna pin
36	GND	Power	Ground
37	GND	Power	Ground
38	UART0_RXD	Digital I/O	UART0 Receive (modem trace)
39	UART0_TXD	Digital I/O	UART0 Transmit (modem trace)
40	GND	Power	Ground
41	GND	Power	Ground
42	VDD	Power	Power Supply
43	VDD	Power	Power Supply
44	GND	Power	Ground
45	GND	Power	Ground
46	GND	Power	Ground
47	GND	Power	Ground
48	GND	Power	Ground
49	GND	Power	Ground
50	GND	Power	Ground
51	GND	Power	Ground
52	GND	Power	Ground
53	GND	Power	Ground
54	GND	Power	Ground
55	GND	Power	Ground
56	GND	Power	Ground
57	RESERVED		Connect thermally and mechanically to the application board but leave electrically unconnected.
58	RESERVED		Connect thermally and mechanically to the application board but leave electrically unconnected.
59	USB_DP	Digital I/O	USB differential data +
60	USB_DN	Digital I/O	USB differential data -
61	USB_VBUS	Digital I/O	USB VBUS input and System OFF wake-up
62	SIM2_CLK	Digital I/O	SIM2 clock
63	SIM2_RST	Digital I/O	SIM2 reset
64	SIM2_IO	Digital I/O	SIM2 data

Pin	Name	Function	Description
65	SIM2_VOUT	Power	SIM power supply
66	UART2_RTS	Digital I/O	UART2 Request To Send
67	UART2_CTS	Digital I/O	UART2 Clear To Send
68	GND	Power	Ground
69	GND	Power	Ground
70	GND	Power	Ground
71	GND	Power	Ground
72	GND	Power	Ground
73	GND	Power	Ground
74	GND	Power	Ground
75	GND	Power	Ground
76	GND	Power	Ground
77	GND	Power	Ground
78	GND	Power	Ground
79	SIM1_DET	Digital I/O	SIM1 detect. If not used it must be left floating.
80	GND	Power	Ground
81	GND	Power	Ground
82	RESERVED		Connect thermally and mechanically to the application board but leave electrically unconnected.
83	GND	Power	Ground
84	GND	Power	Ground
85	GND	Power	Ground
86	GND	Power	Ground
87	GND	Power	Ground
88	GND	Power	Ground
89	GND	Power	Ground
90	GND	Power	Ground
91	GND	Power	Ground
92	GND	Power	Ground
93	GND	Power	Ground
94	GND	Power	Ground
95	GND	Power	Ground
96	GND	Power	Ground

Pin	Name	Function	Description
97	GND	Power	Ground
98	GND	Power	Ground
99	GND	Power	Ground
100	RFC2	Digital I/O	RF control interface
101	RFC3	Digital I/O	RF control interface
102	GND	Power	Ground
103	RESERVED		Connect thermally and mechanically to the application board but leave electrically unconnected.
104	COEX0	Digital I/O	Coexistence and RF control interface
105	COEX1	Digital I/O	Coexistence and RF control interface
106	GND	Power	Ground
107	GND	Power	Ground
108	GND	Power	Ground
109	GND	Power	Ground

Table 13: LGA pin assignments

12.2 Mechanical specifications

Dimensions in millimeters for the nRF93M1 LGA 15.8×17.7×2.4 mm package

12.2.1 Module package

The nRF93M1 uses a 15.8×17.7×2.4 mm LGA package with variable ball pitch. All dimensions are in millimetres and follow JEDEC tolerance conventions unless otherwise stated.

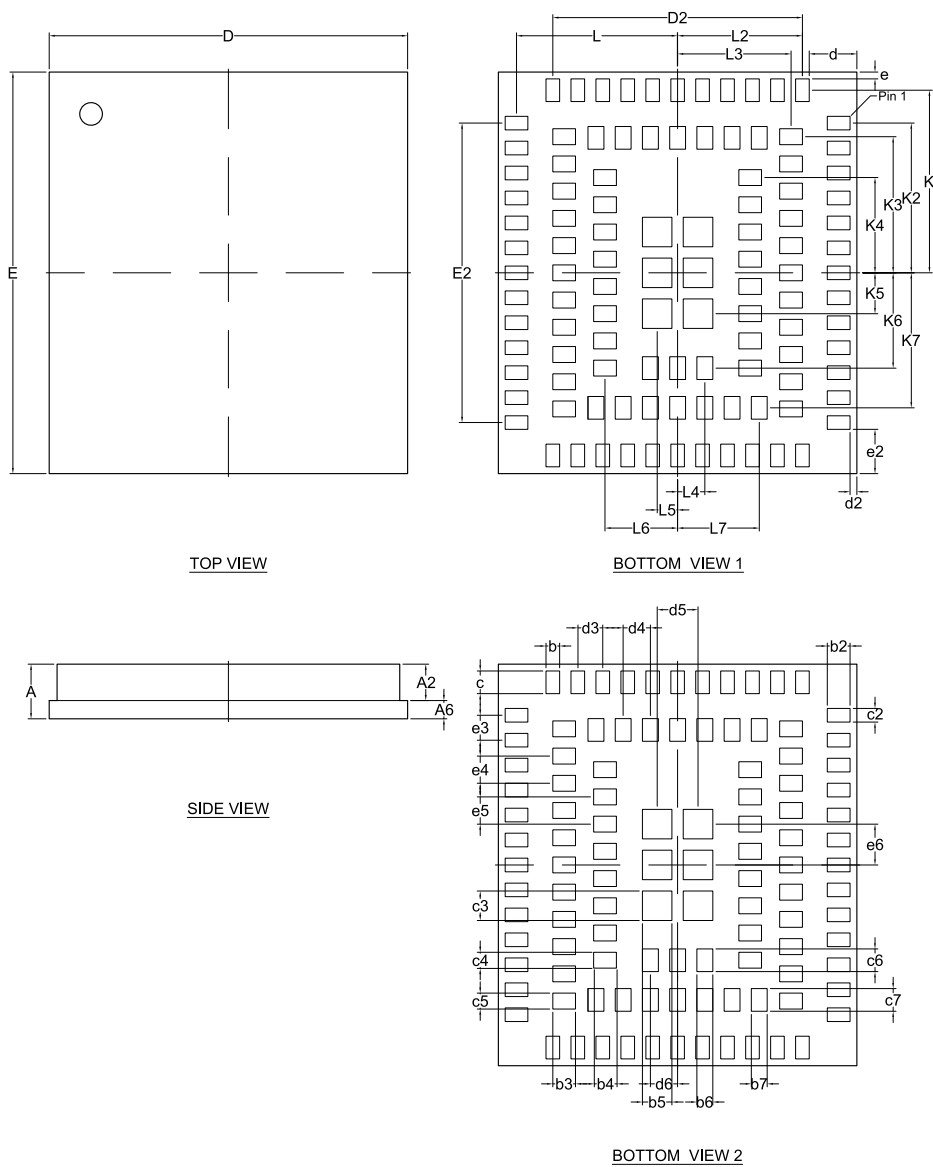


Figure 2: Package dimensions in millimeters

	Min.	Nom.	Max.
A	2.2	2.4	2.6
A2		1.6	
A6		0.8	
b		0.6	
b2		1.0	
b3		1.0	
b4		1.0	
b5		1.3	
b6		0.7	
b7		0.7	
c		1.0	
c2		0.6	
c3		1.3	
c4		0.7	
c5		0.7	
c6		1.0	
c7		1.0	
D	15.65	15.8	15.95
D2		11.0	
d		2.1	
d2		0.3	
d3		1.1	
d4		1.2	
d5		1.8	
d6		1.2	
E	17.55	17.7	17.85
E2		13.2	
e		0.3	
e2		1.95	
e3		1.1	
e4		1.2	
e5		1.2	
e6		1.8	
K		8.05	

	Min.	Nom.	Max.
K2		6.6	
K3		6.0	
K4		4.2	
K5		1.8	
K6		4.2	
K7		6.0	
L		7.1	
L2		5.5	
L3		5	
L4		1.2	
L5		0.9	
L6		3.2	
L7		3.6	

Table 14: Package dimensions in millimeters

Attribute	Value
Package type	LGA, variable ball pitch
Body size (D×E×A)	15.8×17.7×2.4 mm (nominal)
Pin count	109
Weight	1.5 g
RoHS / REACH	Compliant

Table 15: Package attributes

12.3 Reference circuitry

To ensure good RF performance when designing PCBs, using the PCB layouts and component values provided by Nordic Semiconductor is highly recommended.

Documentation for the different package reference circuits, including Altium Designer files, PCB layout files, and PCB production files can be downloaded from the product page at www.nordicsemi.com

This section contains reference circuitry showing the components to support the design of on-chip features.

Note: This is not a complete list of configurations, but all required circuitry is shown for further configurations.

12.3.1 nRF93M1 reference design

For Bill of Materials (BOM), PCB layout and thermal design, see the nRF93M1 Hardware Design Guidelines.

12.4 Thermal characteristics

Thermal data for the nRF93M1 module, including ambient operating range, junction-to-ambient resistance, and guidance for host-PCB thermal design.

The following topic provides thermal parameters for module-level PCB design and thermal analysis.

12.4.1 Thermal characterization

Module-level thermal-resistance and temperature-monitoring data.

Operating temperature range

nRF93M1 operates over an ambient temperature range of -35°C to $+75^{\circ}\text{C}$. Module-level derating (if any) is given in the Recommended Operating Conditions chapter.

Thermal resistance

Parameter	Value	Unit
Junction-to-ambient thermal resistance, θ_{JA}	TBD	$^{\circ}\text{C}/\text{W}$
Junction-to-case thermal resistance, θ_{JC}	TBD	$^{\circ}\text{C}/\text{W}$
Junction-to-top characterization parameter, Ψ_{JT}	TBD	$^{\circ}\text{C}/\text{W}$
Junction-to-board characterization parameter, Ψ_{JB}	TBD	$^{\circ}\text{C}/\text{W}$

Measurements follow JEDEC standards JESD51-2 (θ_{JA} , Ψ_{JT}) and JESD51-8 (θ_{JC}) on a 2s2p test board.

On-chip temperature monitor

The host application can read the on-chip temperature through the proprietary `AT%ADC` command.

Host PCB guidance

For reliable operation at elevated ambient temperature:

- Use a thermal-relief-free copper pour under the module footprint on the host PCB to improve heat spreading.
- Add thermal vias beneath the module when the host PCB uses inner ground planes.
- Avoid placing heat-generating components (high-current regulators, RF power amplifiers, or CPU) directly adjacent to the nRF93M1.

12.5 Assembly-related instructions

nRF93M1 is delivered in a sealed dry pack and follows standard JEDEC handling guidelines for surface-mount LGA modules.

Follow these guidelines when integrating nRF93M1 on the host PCB:

- Storage – keep modules in the sealed dry pack at $\leq 30^{\circ}\text{C}/60\% \text{ RH}$ until ready for assembly.
- Floor life – once the dry pack is opened, complete reflow within 168 hours at $\leq 30^{\circ}\text{C}/60\% \text{ RH}$ (MSL 3).
- Baking – if floor life or storage conditions are exceeded, bake per JEDEC J-STD-033 before reflow.
- Reflow profile – use a profile compatible with IPC-7530, peak $240^{\circ}\text{C} \pm 5^{\circ}\text{C}$, max 3 reflow cycles.

- RESERVED pins – connect thermally and mechanically to the host PCB but leave electrically unconnected.
- Antenna routing – route the ANT pin with a controlled-impedance 50 Ω transmission line; keep the line short and clear of digital aggressors.
- VDD decoupling – place bulk + high-frequency decoupling capacitors close to the VDD pins per the Hardware Design Guidelines.
- ESD handling – observe ESD-protected handling per JESD625 (Class 1C HBM, Class C1 CDM).

For more detailed instructions related to assembly, see nRF93M1 Hardware Design Guidelines.

13 Recommended operating conditions

The operating conditions are the physical parameters that the chip can operate within.

Symbol	Parameter	Notes	Min.	Nom.	Max.	Units
VDD	Module input voltage	Including voltage drop, ripple, and spikes.	3.3	3.8	4.5	V
LDO_OUT	LDO output voltage	Internal LDO output; supplies GPIO pads and is available to the host. 1.8 V only on nRF93M1.	1.71	1.80	1.89	V
SIM1_VOUT/ SIM2_VOUT	SIM output supply voltage (1.8 V mode)	Auto-selected from UICC class. Supplied by internal LDO.	1.71	1.80	1.89	V
SIM1_VOUT/ SIM2_VOUT	SIM output supply voltage (3 V mode)	Auto-selected from UICC class. Supplied by internal LDO.	2.85	3.00	3.15	V
T _A	Operating ambient temperature		-35	25	+75	°C

Table 16: Recommended operating conditions

Note: There can be excessive leakage at VDD and/or LDO_OUT if any of these supply voltages is outside its range given in the table above.

13.1 LDO_OUT Considerations

LDO_OUT is the internal LDO output that supplies the GPIO pads and is available to the host as an auxiliary low-current rail.

The following restrictions should be taken into consideration:

- LDO_OUT is derived from VDD; ensure VDD is applied before using LDO_OUT.
- LDO_OUT should be considered de-energised before removing VDD.
- LDO_OUT is not available in System OFF.
- If VDD is supplied and LDO_OUT is grounded externally, extra current consumption can be generated on VDD.

14 Absolute maximum ratings

Maximum ratings are the extreme limits to which the chip can be exposed for a limited amount of time without permanently damaging it. Exposure to absolute maximum ratings for prolonged periods of time may affect the reliability of the device.

Note	Min.	Max.	Unit
VDD	-0.3	5.0	V
SIM_IO, SIM_CLK, SIM_RST, SIM_DET	-0.3	3.6	V
GND		0	V

Table 17: Supply voltage

Note	Min.	Max.	Unit
ANT input level		10	dBm
RF port ruggedness	Maximum deviation from 50 Ω without damaging the module	10:1	VSWR

Table 18: Radio

Note	Min.	Max.	Unit
Storage temperature	-40	90	°C
Reflow soldering temperature		245	°C
MSL	Moisture Sensitivity Level	3	
ESD HBM	Human Body Model	1.5	kV
ESD HBM Class	Human Body Model Class	1C	
ESD CDM	Charged Device Model	350 V	V
ESD CDM Class	Charged Device Model Class	C1	

Table 19: Environmental (LGA package)

Note	Min.	Max.	Unit
Endurance	100,000		Write/erase cycles
Retention	10 years at 75°C		y

Table 20: Flash memory

15 Ordering information

This chapter contains information on module marking, ordering codes, and container sizes.

15.1 LGA marking

The nRF93M1 package is laser-marked on the top side with vendor identification, product name, build code, and traceability information.

The detailed mapping between marking fields and the order code grammar is given in [Code ranges and values](#) on page 46.

15.2 Order code

The nRF93M1 order code identifies the product family, package, function variant, build code, and shipping container.

The order code has the following structure:

nRF93M1-LA<xA>-<CC>

Each field of the order code is defined in the table below. The associated build-code and serial-number fields, used in package marking and shipping-container labels, are defined in [Code ranges and values](#) on page 46.

Abbreviation	Definition and implemented codes
nRF93	nRF93 Series product
M1	Module variant
LA	Package variant code (LGA)
<xA>	Function variant code (band/region option — see Ordering options on page 48)
<H><P><F>	Build code: H hardware version, P production configuration, F firmware version (only visible on shipping container label)
<YY><WW><LL> <VC><XXXXXX><ZZ>	Serial number: year, assembly week, lot code Vendor code, alphanumeric serial number, checksum
<CC>	Container code (reel size — see Code ranges and values on page 46)

Table 21: Order code abbreviations

15.3 Code ranges and values

This topic defines the value ranges used in the nRF93M1 order code, build code, and serial number.

<H>	Description
[A . . Z]	Hardware version/revision identifier (incremental)

Table 22: Hardware version codes

<P>	Description
[0 . . 9]	Production device identifier (incremental)
[A . . Z]	Engineering device identifier (incremental)

Table 23: Production configuration codes

<F>	Description
[A . . N, P . . Z]	Version of preprogrammed firmware
[0]	Delivered without preprogrammed firmware

Table 24: Firmware version codes

<YY>	Description
[25 . . 99]	Production year: 2025 to 2099

Table 25: Year codes

<WW>	Description
[01 . . 53]	Week of production

Table 26: Week codes

<LL>	Description
[AA . . ZZ]	Production lot identifier

Table 27: Lot codes

<VC>	Description
[LQ]	Module manufacturer

Table 28: Vendor code

<XXXXXX>	Description
[00000 . . ZZZZZ]	Alphanumeric serial number

Table 29: Serial number

<ZZ>	Description
[00 . . ZZ]	Alphanumeric serial-number checksum

Table 30: Serial number checksum

<CC>	Description
R7	7" reel
R	13" reel

Table 31: Container codes

15.4 Ordering options

nRF93M1 is offered in multi-regional and global LTE-band variants on 7" and 13" reels.

The available nRF93M1 order codes and the corresponding minimum ordering quantities are listed in the following table.

Order code	Minimum ordering quantity (MOQ)	Comment
nRF93M1-LABA-R	1 000	Multi-regional band option, 13" reel
nRF93M1-LABA-R7	100	Multi-regional band option, 7" reel
nRF93M1-LACA-R	1 000	Global band option, 13" reel
nRF93M1-LACA-R7	100	Global band option, 7" reel

Table 32: nRF93M1 order codes

16 Regulatory information

This module and the Development Kit for the module have been tested and comply with RF exposure requirements at a minimum separation distance of 20 cm from the human body, with no co-location with other transmitters.

17 Legal notices

By using this documentation you agree to our terms and conditions of use. Nordic Semiconductor may change these terms and conditions at any time without notice.

Liability disclaimer

Nordic Semiconductor ASA reserves the right to make changes without further notice to the product to improve reliability, function, or design. Nordic Semiconductor ASA does not assume any liability arising out of the application or use of any product or circuits described herein.

Nordic Semiconductor ASA does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. If there are any discrepancies, ambiguities or conflicts in Nordic Semiconductor's documentation, the Datasheet prevails.

Nordic Semiconductor ASA reserves the right to make corrections, enhancements, and other changes to this document without notice.

Customer represents that, with respect to its applications, it has all the necessary expertise to create and implement safeguards that anticipate dangerous consequences of failures, monitor failures and their consequences, and lessen the likelihood of failures that might cause harm, and to take appropriate remedial actions.

Nordic Semiconductor ASA assumes no liability for applications assistance or the design of customers' products. Customers are solely responsible for the design, validation, and testing of its applications as well as for compliance with all legal, regulatory, and safety-related requirements concerning its applications.

Nordic Semiconductor ASA's products are not designed for use in life-critical medical equipment, support appliances, devices, or systems where malfunction of Nordic Semiconductor ASA's products can reasonably be expected to result in personal injury. Customer may not use any Nordic Semiconductor ASA's products in life-critical medical equipment unless adequate design and operating safeguards by customer's authorized officers have been made. Customer agrees that prior to using or distributing any life-critical medical equipment that include Nordic Semiconductor ASA's products, customer will thoroughly test such systems and the functionality of such products as used in such systems.

Customer will fully indemnify Nordic Semiconductor ASA and its representatives against any damages, costs, losses, and/or liabilities arising out of customer's non-compliance with this section.

RoHS and REACH statement

Refer to www.nordicsemi.com for complete hazardous substance reports, material composition reports, and latest version of Nordic's RoHS and REACH statements.

Trademarks

All trademarks, service marks, trade names, product names, and logos appearing in this documentation are the property of their respective owners.

Copyright notice

© 2026 Nordic Semiconductor ASA. All rights are reserved. Reproduction in whole or in part is prohibited without the prior written permission of the copyright holder.



Contents

	nRF93M1 Feature overview.	ii
1	Revision history.	3
2	About this document.	4
	2.1 Document status.	4
3	Product overview.	5
	3.1 Block diagram.	5
	3.2 Feature matrix.	6
4	Power and clock management.	9
	4.1 Power supply system.	9
	4.1.1 Power On and Off.	9
	4.1.2 I/O power supply.	10
	4.1.3 LDO output.	10
	4.2 Power supply and temperature monitoring.	10
	4.3 System power modes.	11
	4.3.1 System ON.	12
	4.3.2 System OFF.	12
	4.3.3 System Disabled.	12
	4.4 Power saving.	13
	4.4.1 Automatic power saving in System ON.	13
	4.4.2 Host interface signaling DTR and RI.	13
	4.4.3 Power Saving Mode (PSM).	14
	4.4.4 Extended DRX (eDRX).	14
	4.4.5 IDLE DRX and connected DRX.	14
	4.4.6 Modem functionality levels.	15
	4.5 Host configuration.	15
	4.5.1 Recommended power saving configuration.	16
	4.6 Reset.	17
	4.7 Current consumption.	18
5	LTE Modem and connectivity stack.	20
	5.1 Supported bands and frequencies.	20
	5.2 Connectivity stacks.	22
	5.3 SMS.	22
	5.4 Firmware update.	23
	5.5 AT command interface.	24
	5.6 SIM Interfaces.	24
	5.7 RF control interface.	24
	5.8 LTE coexistence interface.	25
	5.9 RF front-end interface.	25
	5.10 RF electrical specification.	25
6	Wi-Fi Location.	26
7	Cellular Location.	27

8	Device interfaces.	28
8.1	Status and Network indicators.	28
8.1.1	STATUS indicator.	28
8.1.2	NETWORK indicator.	28
8.2	USB.	28
8.3	UART.	28
8.3.1	UART1 – Primary.	29
8.3.2	UART0 – Trace.	29
8.3.3	UART2 – Auxiliary.	29
9	Security.	30
10	Debug and trace.	31
11	nRF Cloud services.	32
12	Hardware and layout.	33
12.1	Pin assignments.	33
12.1.1	Module pin assignments.	33
12.2	Mechanical specifications.	37
12.2.1	Module package.	37
12.3	Reference circuitry.	40
12.3.1	nRF93M1 reference design.	40
12.4	Thermal characteristics.	41
12.4.1	Thermal characterization.	41
12.5	Assembly-related instructions.	41
13	Recommended operating conditions.	43
13.1	LDO_OUT Considerations.	43
14	Absolute maximum ratings.	44
15	Ordering information.	46
15.1	LGA marking.	46
15.2	Order code.	46
15.3	Code ranges and values.	46
15.4	Ordering options.	48
16	Regulatory information.	49
17	Legal notices.	50